SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECHNOLOGY, SURAT
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CS103: DESIGN AND ANALYSIS OF ALGORITHMS
M.Tech. (DS) Ird Year
Lab Assignment

Assignment 1

Suppose a consulting firm that offers security consultancy needs to obtain licenses for n different pieces of cryptographic software. Due to regulations, they can buy only one license per month. Each license currently costs Rs 1000, but the cost of each license increases at an exponential rate $r_j (> 1)$ every month. For example, if license j is purchased t months from now, it will cost $100 \times r_j^t$. All price growth rates are distinct. Given the rates of price growth, determine the order in which licenses should be bought to minimize total cost. Implement this in a program.

Assignment 2

An Euler circuit in a directed graph is a cycle in which every edge is visited exactly once. Design and implement a linear-time algorithm to find an Euler circuit in a directed graph (where one exists).

Assignment 3

You are given a set of N sticks, each specified by its two endpoints as ordered triples (x, y, z). No stick is vertical, and a stick can only be picked up if no other sticks are lying on top of it.
(a) Write a routine that takes two sticks a and b and reports whether a is above, below, or unrelated to b.
(b) Give an algorithm that determines whether it is possible to pick up all the sticks, and if so, provides a sequence of stick pickups that accomplishes this.

Assignment 4

A graph is k-colorable if each vertex can be given one of k colours and no edge connects identically coloured vertices. Give a linear-time algorithm to test a graph for two-colorability. Assume graphs are stored in adjacency-list format and specify any additional data structures that are needed. How is a two-colorable graph related to a bipartite graph? Provide at least two applications of two-colorability or bipartite graphs.

Assignment 5

Edge-weighted graphs can be interpreted as a network of pipes, where the weight of edge (i, j) determines the pipe's capacity. The network flow problem asks for the maximum flow from a source vertex s to a target vertex t in a weighted graph G, respecting the capacity of each pipe.
• Create a source node s connected to every vertex in set L by an edge of weight 1.
• Create a sink node t and connect it to every vertex in set R by an edge of weight 1.
• Assign each edge in the bipartite graph G a weight of 1.
Now, the maximum possible flow from s to t defines the largest matching in G. Implement the algorithm.

Assignment 6

A person wants to fly from city A to city B. Suppose several airlines have connections in different cities, and the flights operate on different schedules. The goal is to minimize travel time or cost. Write a program that implements the shortest-path algorithm to help the person make the most efficient travel arrangements.